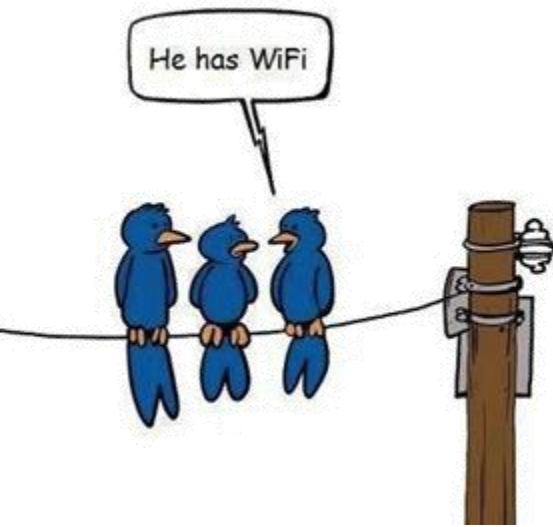


# ECSE 4560/ 6560

## Modern Communication Systems

### Lecture 1: Introduction



Ish Jain, Fall 2025



Slides courtesy of Haitham Al  
Hassanieh from EPFL

Increasing  
Demand for  
Wireless  
Connectivity

Connecting  
People



Increasing  
Demand for  
Wireless  
Connectivity



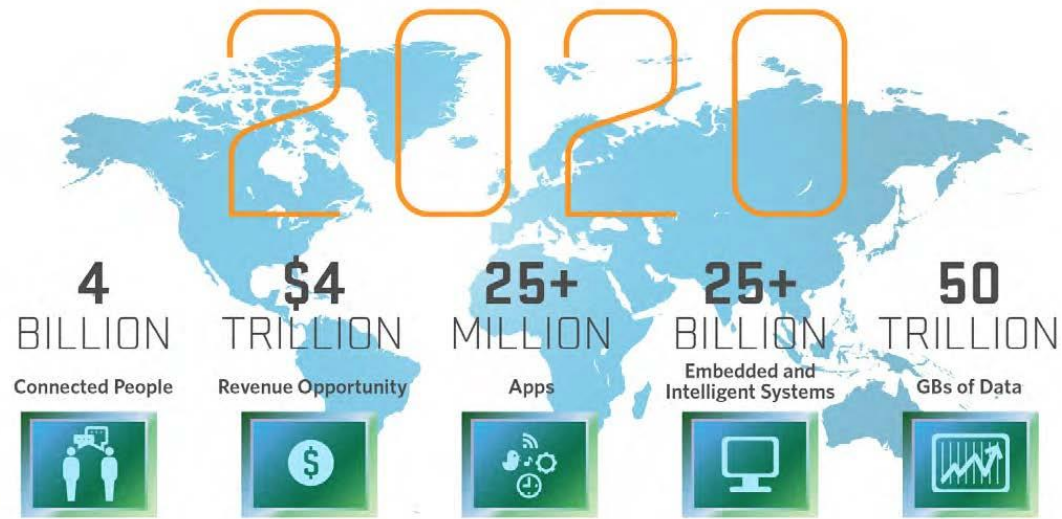
Connecting  
Everything



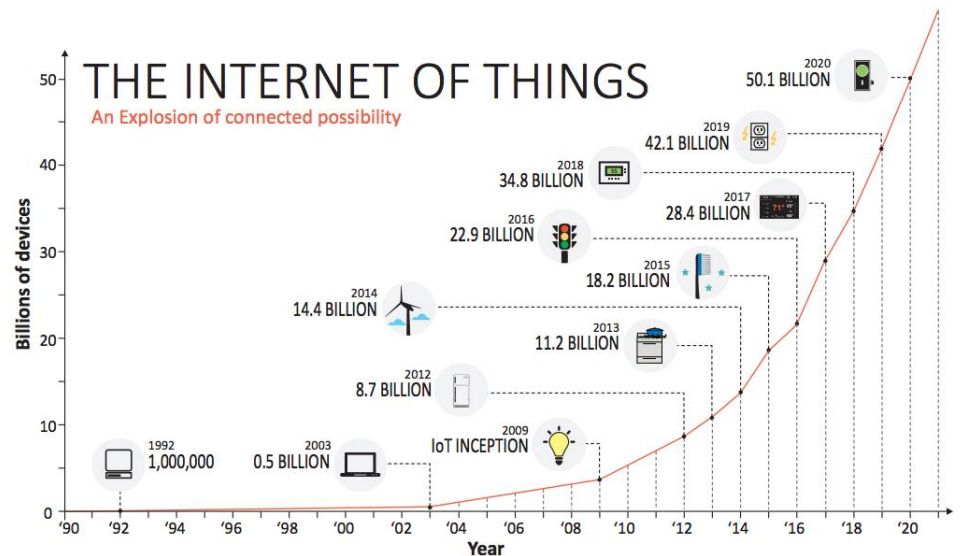
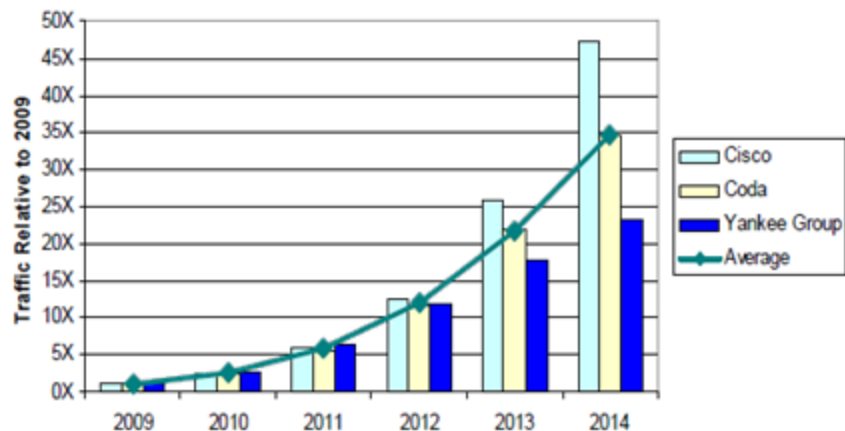




# Increasing Demand for Wireless/IoT Connectivity



Source: Mario Morales, IDC.

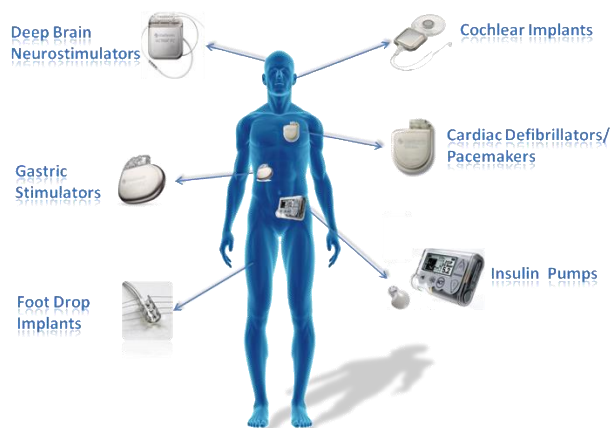


# Wireless Sensing and Communication

## Wireless Homes



## Wireless Biomedical Implants



## Wireless Wearables



## Cellular Networks



## Wireless Sensors



## UAVs



## Wireless Data Centers



## Wireless VR



## Wireless Vehicles





# Wireless for Control and Sensing

All interaction from objects, machines and humans



Self-driving cars



Robots



AR, VR



IoT sensors

...a network to accommodate these devices

ECSE 4660/6660: Internetworking of Things – **Spring 2025**

ECSE 4560/6560: Modern Communication Systems – **Fall 2025**

# Administrative Information

## ■ Class meeting

- Lecture Time: Tu, Fr 10:00-11:20 AM
- Location (in-person only):
  - JEC 4104
- Lecture on 8/28, 9/7 may be rescheduled/delivered online
- Lab: JEC 5312 (access will be provided)

## ■ Instructor

- Prof. Ish Jain
  - Office: JEC 6018
  - Email: [jaini@rpi.edu](mailto:jaini@rpi.edu)
- Office Hours: Tu, Fr after class or by appointment

## ■ TA

- Shukai Chen
  - Email: [chens37@rpi.edu](mailto:chens37@rpi.edu)
- Office Hours: TBD (likely Friday- HW submission day)



# Web Resources

- Piazza “Resources” for Course Material
  - Lecture Slides
  - Reading Lists
  - Homeworks + Matlab Assignments
- LMS
  - Important announcement and Per-class Quiz
- Piazza “Q&A” for Discussions
  - Annoucements
  - Discussions, Q&A
- Join Piazza now:  
<https://piazza.com/rpi/fall2025/ecse45606560>
- GradeScope for assignment submission (You will be added)

# Textbook

- *Required:*

- A. Goldsmith (2005). Wireless Communications, Cambridge University Press. (available as e-book through RPI library)

- *References:*

- Carlson and Crilly (2009), Communication Systems (5th ed.). McGraw Hill.
- D. Tse and P. Viswanath (2005). Fundamentals of Wireless Communication, Cambridge University Press. (available as e-book through RPI library)
- J. Proakis and M. Salehi (2008). Digital Communications (5th ed.). McGraw Hill.
- A. F. Molisch (2011). Wireless Communications (2nd ed.). Wiley Publishing.

# Course Content

| Lecture # | Topic                                                          |
|-----------|----------------------------------------------------------------|
| 1         | Intro to wireless communication and sensing systems            |
| 2,3       | Signals review, channel modeling, modulation schemes           |
| 4,5       | Communication Architecture                                     |
| 6,7,8,9   | Multi-carrier modulation and OFDM                              |
| 10,11     | Multi-antenna systems and MIMO, mmWave beamforming             |
| 12,13     | Multi-user systems (TDMA, FDMA, CDMA, OFDMA)                   |
| 14        | Exam 1                                                         |
| 15,16     | Fundamentals of Sensing: Space Fourier Transform               |
| 17,18     | Monostatic Radar FMCW sensing                                  |
| 19, 20,21 | Bistatic Sensing – localization and tracking with WiFi         |
| 22,23,24  | Cellular communication standards (3G, 4G/LTE, 5G, 6G) and WiFi |
| 25        | Exam 2                                                         |
| 26,27,28  | Other Emerging Topics and Final project presentations          |



# Course Assessment

- Attendance and In-class participation
  - 5% of the grades
- Per-class Quizzes on LMS
  - 5% of the grades
- Homeworks and Lab (Matlab) Assignments
  - ~8 Homeworks/ Lab Assignments
  - 40% of the overall grade
- 2 midterm exams
  - Dates TBA
  - 30% of the grade (20%, 10% each exam) for 4560 students
  - 20% of the grade (10%, 10% each exam) for 6560 students
- Final Project
  - 20% of the grade for 4560 students
  - 30% of the grade for 6560 students

# Lectures and Quizzes

- Will post a recorded video of each lecture
- A Quiz for each lecture
  - Due end of class day
    - Auto graded on LMS. Beyond midnight, it will be marked late and will not be graded.
  - The quiz will check your knowledge of the material discussed in class
  - Simple questions that can be answered just by attending
  - Questions that require more thinking and studying will be deferred to the homeworks

# Final Project

- Individual or 2 students per group
  - Group formation due Week 2
- Project Proposal
  - Proposal due in Week 4, needs to be approved by Instructor
- Mid-tern Report
  - Due Week 8
- Final Report, In-class presentation, Peer-review
  - Last week of classes
- Prefer students team up with same level (4560 or 6560)
- 4560 Students: Read two papers and summarize (paper list will be provided)
- 6560 Students:
  - Independently replicate results from a paper (Theory/ Simulation) or practically build a system (e.g. build hardware prototype)
  - You may utilize IoT lab resources + you may ask for additional material for project, e.g.,
    - Mobile Robots
    - USRP
    - PlutoSDR
    - Millimeter-wave Radars

# Standard Policy

## ■ Academic Honesty

- Please read about what constitutes academic dishonesty e.g.
  - Failure to cite references
  - Making up results
  - Copying problem solutions/schemes but presenting them differently
  - Presenting work from another course to get credit for multiple courses at once
  - Others
- Penalty for dishonesty
  - Depends on severity and frequency, but could vary including any or all of the following:
  - 0 score on the assignment for 1st instance + reporting to Dean
  - F in the course + reporting to the dean

## ■ Late Policy

- Late submission of assignments are generally not allowed, and late submission without prior instructor permission will not be graded.
- Exception-1: Submissions are typically due on Friday and a late submission until Monday will incur a 20% penalty.
- Exception-2: You are allowed a total of 2 late submissions of your choice until Monday free of penalty.
- Unexpected events (e.g. sickness or accidents) must be supported by a letter from the dean of students.



# In this Class

## Wireless Networks

## Wireless Sensing

### 5G & Next Generation



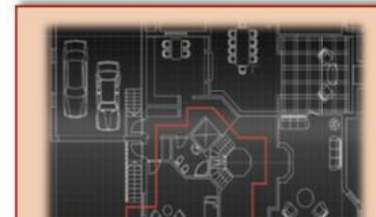
- Millimeter Wave Networks
- Massive MIMO
- Full Duplex Radios
- Dynamic Spectrum Access
- Programmable Surfaces

### Internet of Things



- LoRa Networks
- Backscatter Networks
- Smart Cities & Homes
- Acoustic IoT
- IoT security

### Wireless Localization



- WiFi Localization
- Battery Free Localization
- Antenna Arrays
- Time of Flight
- Tracking

### Wireless Imaging



- Radar Imaging
- Self Driving Cars
- Through wall Imaging
- Human Imaging
- 3D Mesh Recovery

### Cross Layer Wireless



- Rateless Code & Soft PHY
- Interference Alignment
- Virtual MIMO
- Opportunistic Routing
- Network Coding

### Emerging Areas



- Self-driving Cars
- Medical Micro-Implants
- Robotics & Drones
- Underwater Networks
- Noise Cancellation

### Wireless Sensing



- Gesture Sensing
- Vital Sign Monitoring
- Behavioral Sensing
- Human Sensing
- Food Sensing

# Wireless Networks

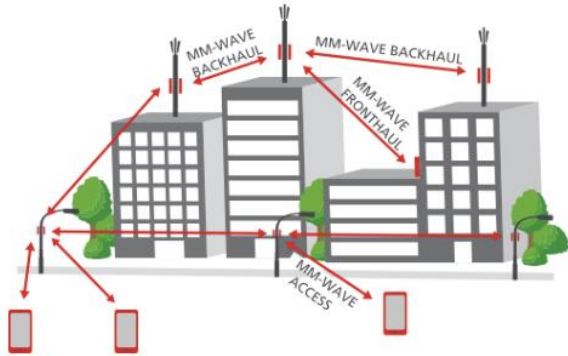
## Wireless networks provide advantages

- Mobility
- Eliminates piles of wires at home and office

## But wireless networks present different challenges

- Medium is shared → **Interference**
- Medium is shared → **Throughput** ↘ as **Devices** ↗
- Channel can be bad & unstable → **Losses + Dead Spots**

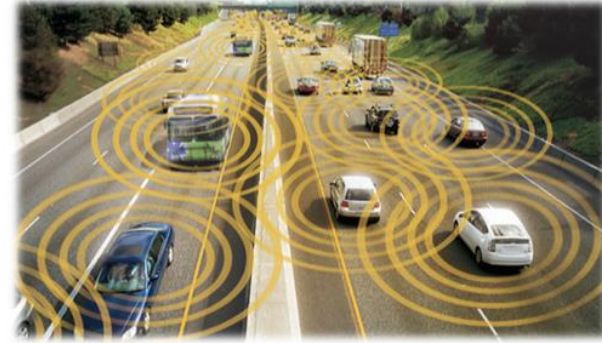
# High Data Rate Applications



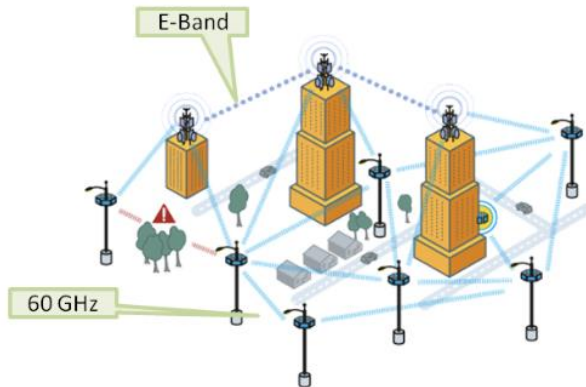
5G Wireless Backhaul & Access



Virtual Reality



Connected Vehicles



60 GHz WiGi Mesh Networks



Robotic Networks



Wireless Data Centers

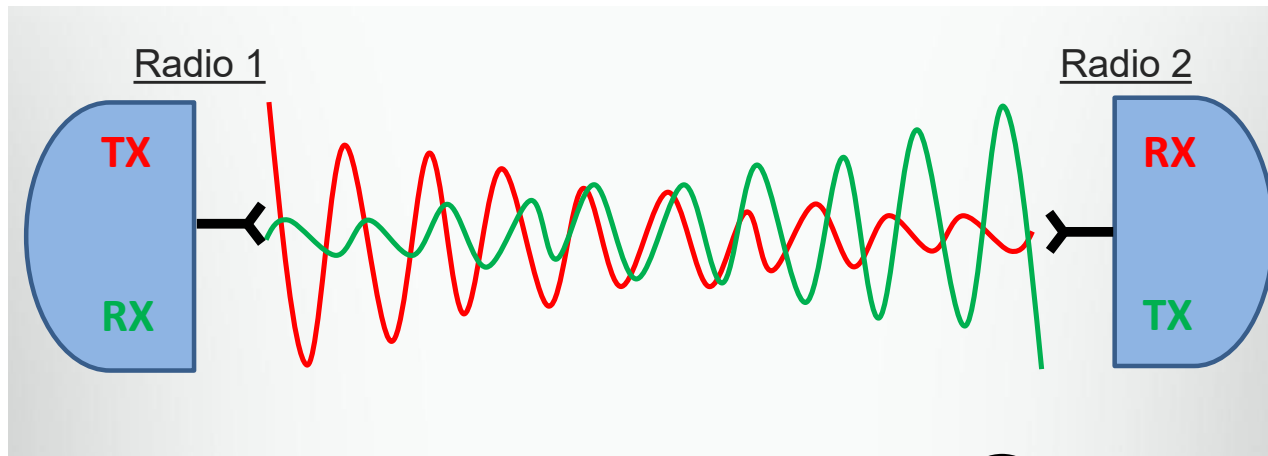


# VR requires a cable connection to a PC



Significantly limits mobility & experience

# Today's Radios Are Half Duplex



Self Interference is 110dB+ stronger than the received signal!



- (A) 100 times
- (B) 100 Thousand times
- (C) 100 Million times
- (D) 100 Billion times

**But we know the signal which we are transmitting!** ✓

→ Cancel the self-interference on the hardware

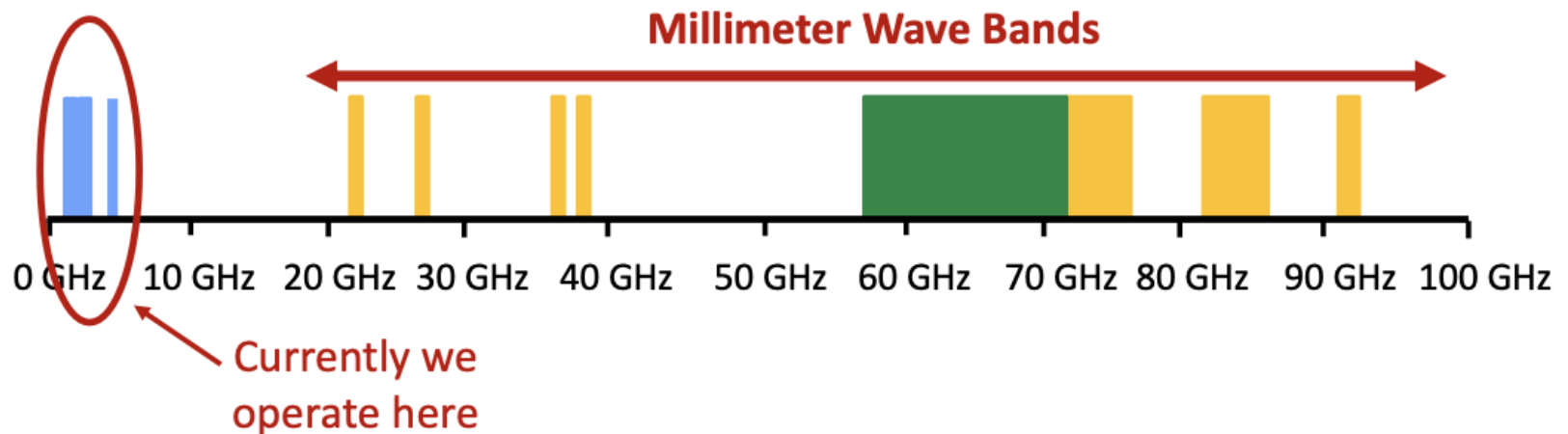
→ 1.97x increase in throughput

**Full Duplex Radios: Major change in communication protocols**



# Millimeter Wave Technology

Huge bandwidth available at millimeter wave frequencies



Millimeter Wave can support data rates of multi-Gbps

# Millimeter Waves Suffer from Attenuation

mmWave radios use phased antenna arrays to focus the power along one direction



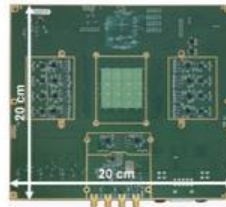
Nokia & National  
Instruments



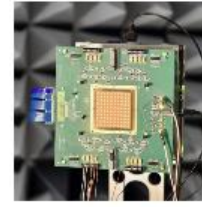
UCSD  
256 elements



UCSD  
64 elements



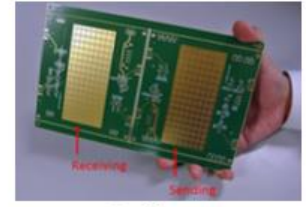
Bell Labs  
384 elements



Anokiwave  
256 elements



IBM  
64 elements



Fujitsu  
64 elements

Small Wavelength enables thousands of antennas to be packed into small space → Extremely narrow beams

## Traditional Approach

Optimize within isolated layers

Network & Apps

Comms. and Coding

HW and Radios

Disruptive gains are unlikely

## New Approach

Optimize across the layers

Network and Apps

Comms. and Coding

HW and Radios

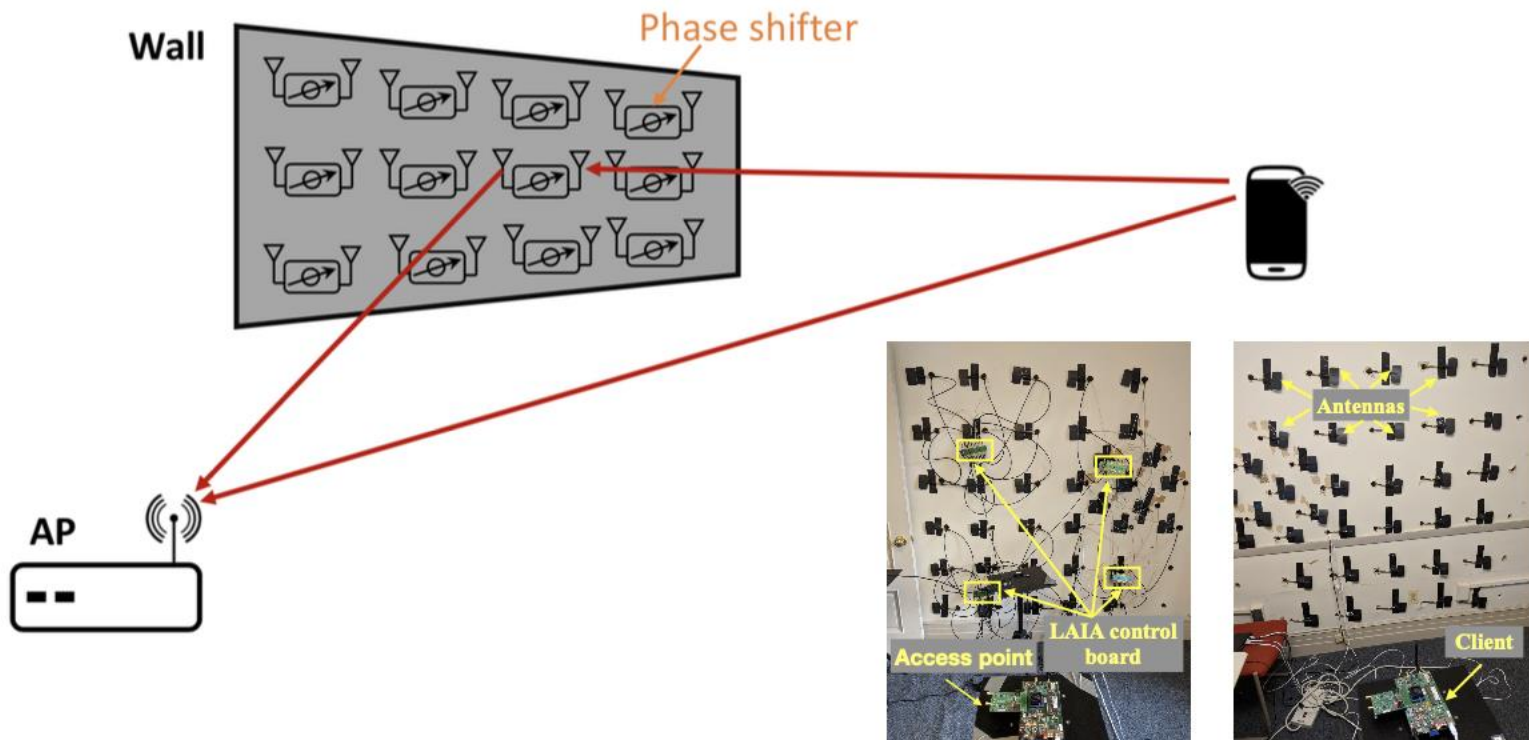
Major opportunities!

Cross-Layer Designs

Fundamental Change in Network Architecture

# Today: Wireless Communication Adapt to Channel Conditions

## New Approach: Programmable Radio Surfaces Change the wireless channel itself



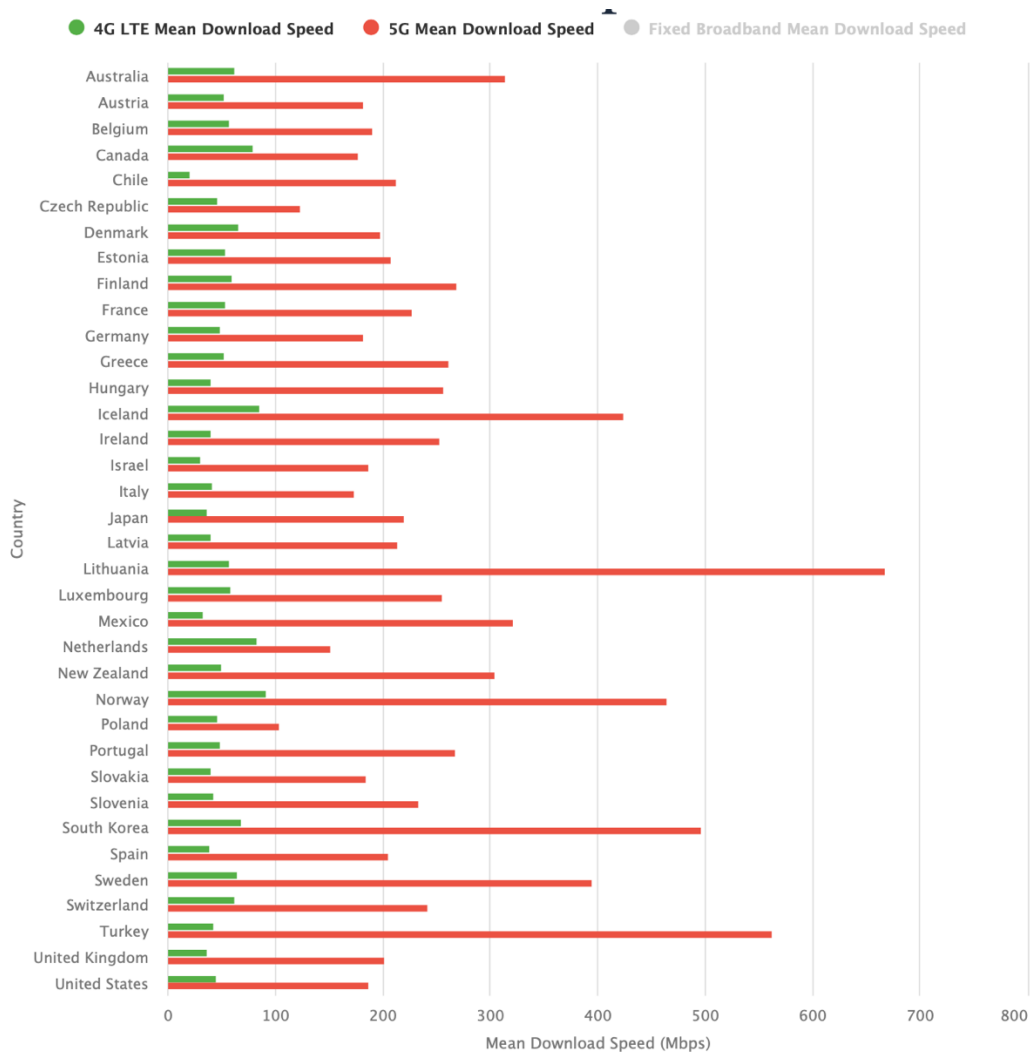
# History of Wireless

## ■ Brief timeline

- 1860-1890: theory and experiments on electromagnetic wave propagation – Maxwell and Hertz
- 1890-1905: First experiments for wireless information transmission  
– Tesla, Bose, Marconi
- 1905-1945: First systems, and use in World Wars I and II
- 1948-1960: Fundamentals of information theory worked out
- 1960s: First satellites launched
- 1960-1980: Initial years of the Internet (ARPANET)
- 1980 onwards: Cellular technology and WiFi
- 2000 onwards: Internet of things
- Emerging applications (2010 and beyond): Satellite communications, Body-area and biochemical communications, quantum communications etc.



# 4G/5G speeds across the globe



[Fig src: <https://www.fcc.gov/2022-communications-marketplace-report-quick-facts>]

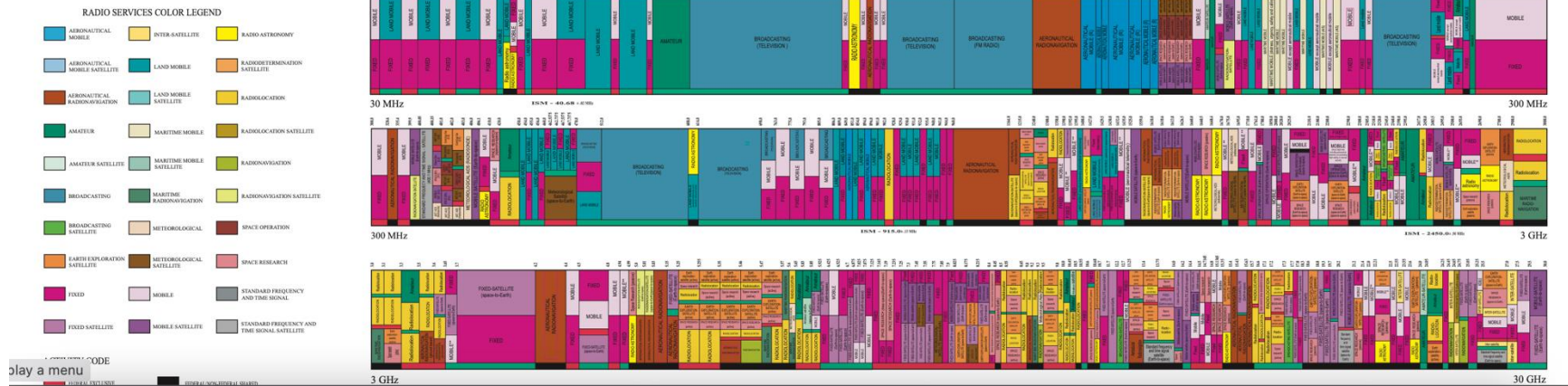
# Spectrum allocation and regulation

- Spectrum a scarce public resource, hence allocated
- Unauthorized spectrum use needs to be monitored/regulated
- Spectral allocation in US controlled by FCC (commercial) or OSM (defense)
- Worldwide spectrum allocation controlled by ITU-R
- Some spectrum set aside for universal use (e.g., research)
- See next slide for frequency allocation map in US

# FCC spectrum allocations

## UNITED STATES FREQUENCY ALLOCATIONS

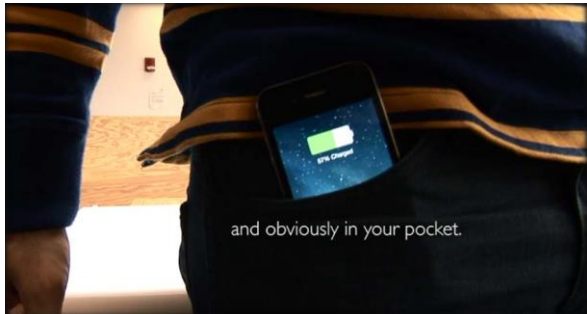
### THE RADIO SPECTRUM



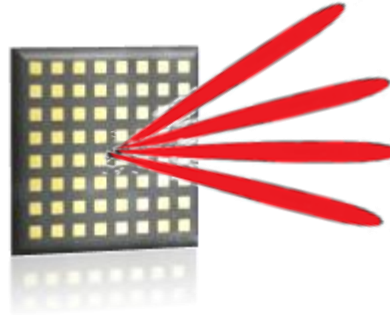
[Fig src: [https://www.ntia.gov/sites/default/files/publications/january\\_2016\\_spectrum\\_wall\\_chart\\_0.pdf](https://www.ntia.gov/sites/default/files/publications/january_2016_spectrum_wall_chart_0.pdf)]

# Other Communication Scenarios

## Wireless Charging

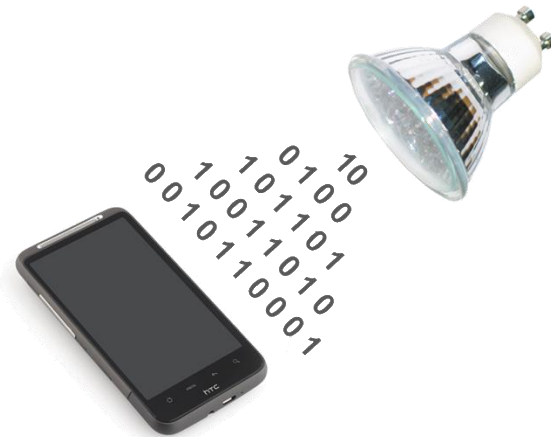


## Millimeter Wave Communication



Wireless at Fiber  
Optic Speed using  
massive antenna  
arrays

## Visible Light Communication



# New Services: Wireless Localization

GPS does not work indoor → Use WiFi to localize.



Indoor Navigation



Business Analytics



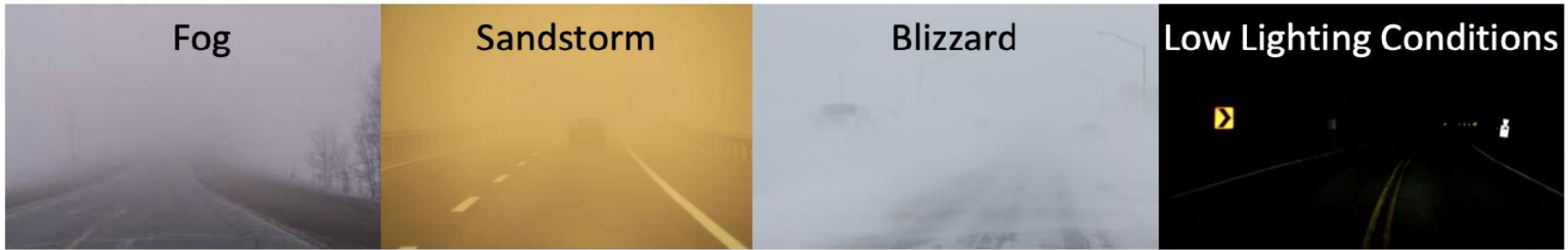
WiFi Geofencing



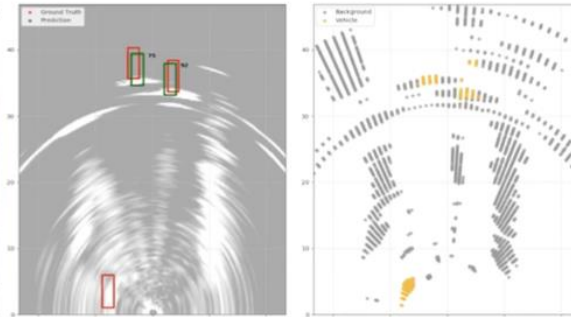
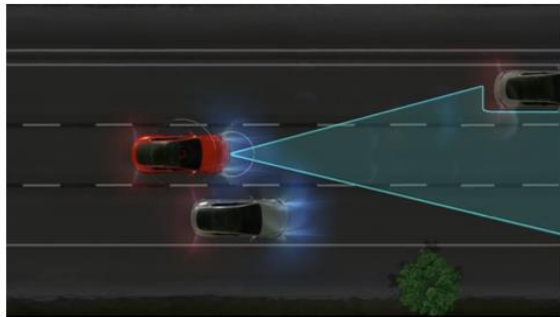
Indoor Robotic Navigation



# Millimeter Wave radar can function in adverse conditions



## State of the Art: Automotive Radars only used for 1D/2D Ranging



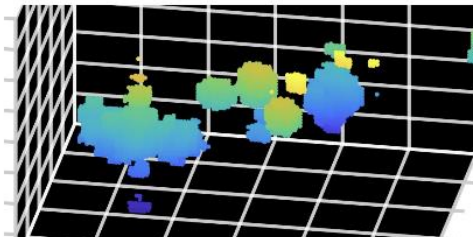
Qualcomm  
snapdragon ride  
platform



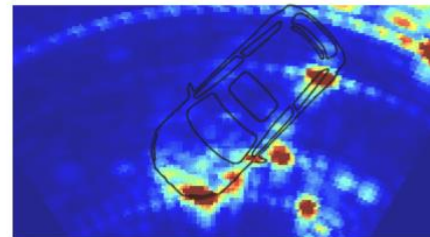
## Radar Imaging is Low Resolution & Suffers from Specularity & Artifacts



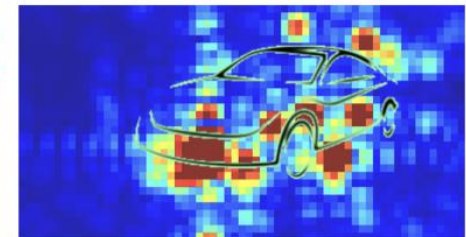
(a) Camera Image



(b) Radar Point Cloud



(c) Top-View Heatmap

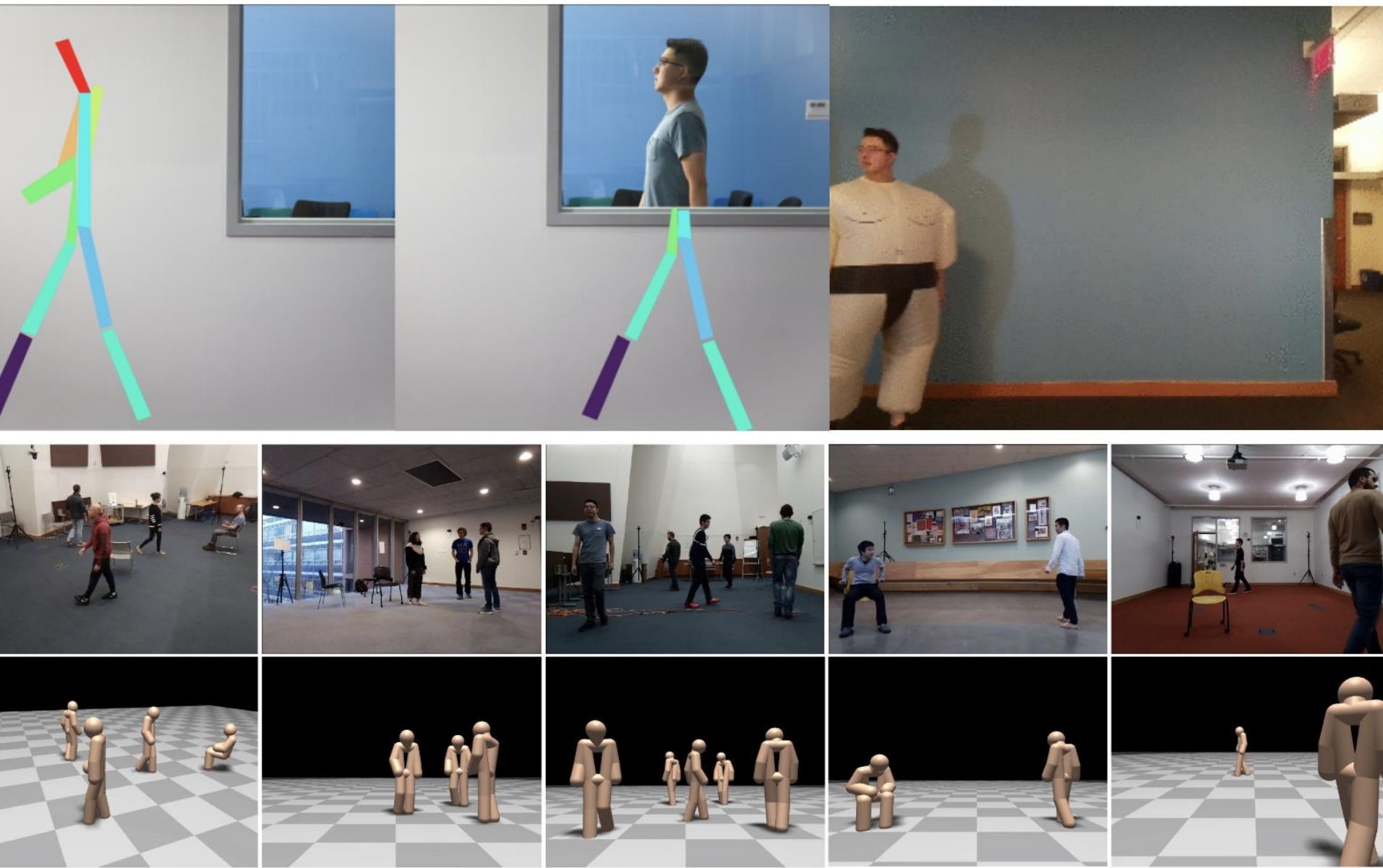


(d) Front-View Heatmap

# Self-Driving Cars in Fog: LiDAR & Camera Fail



# Use Deep Learning to Enhance RF Imaging



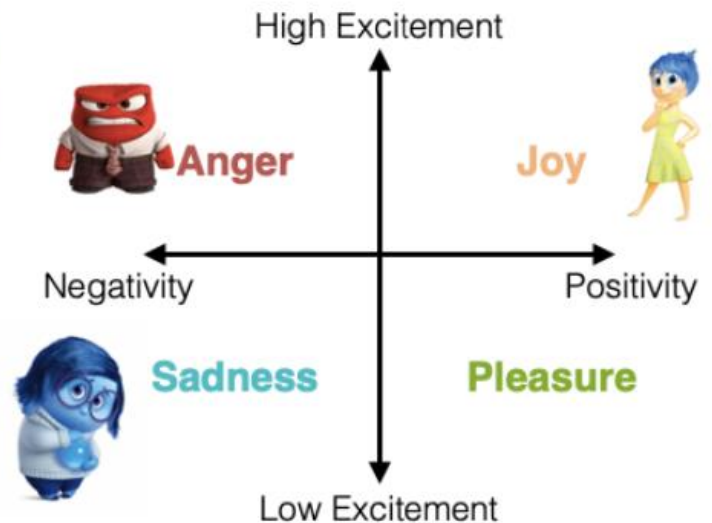
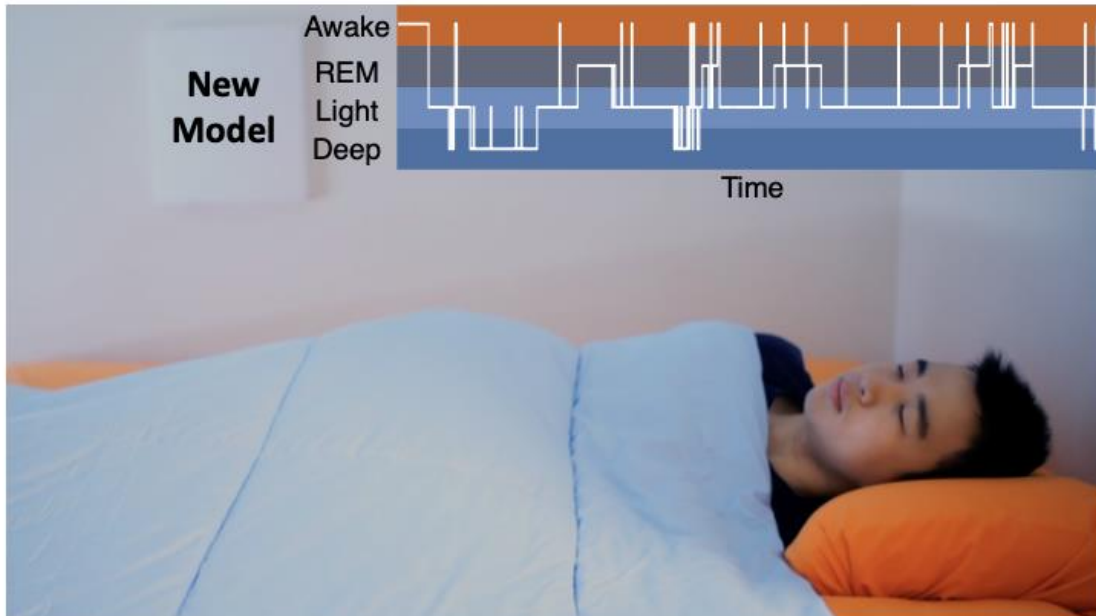
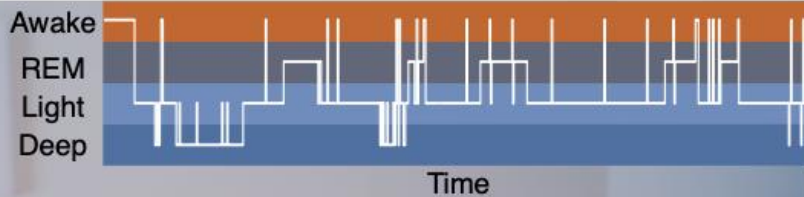


# Sensing Emotions & Sleep Stages



RF signals reflect off body and change with physiological signals

**New  
Model**



# New Services: Wireless Localization

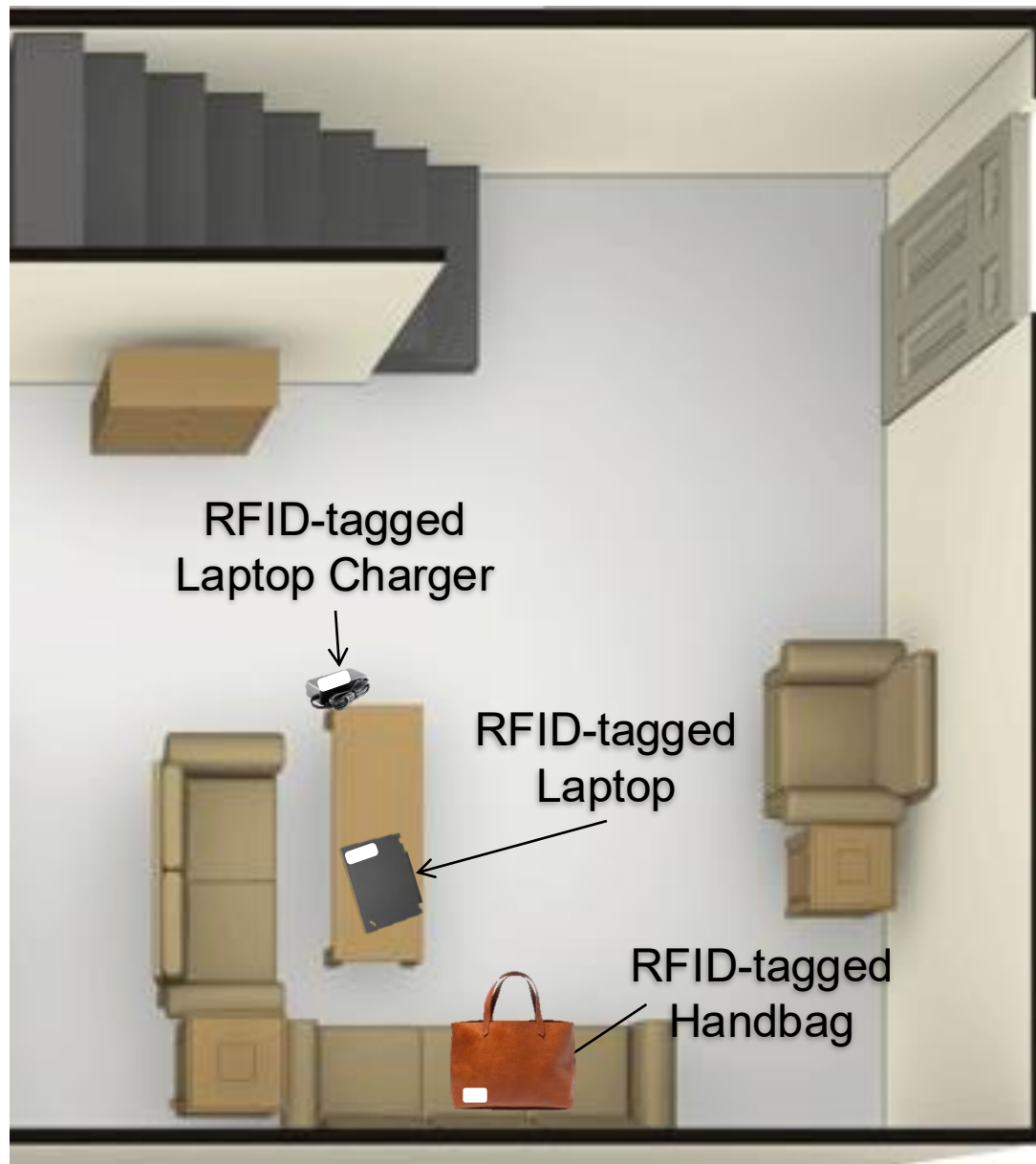
Localize Everything and Anything!



Battery-free stickers to tag any and every object



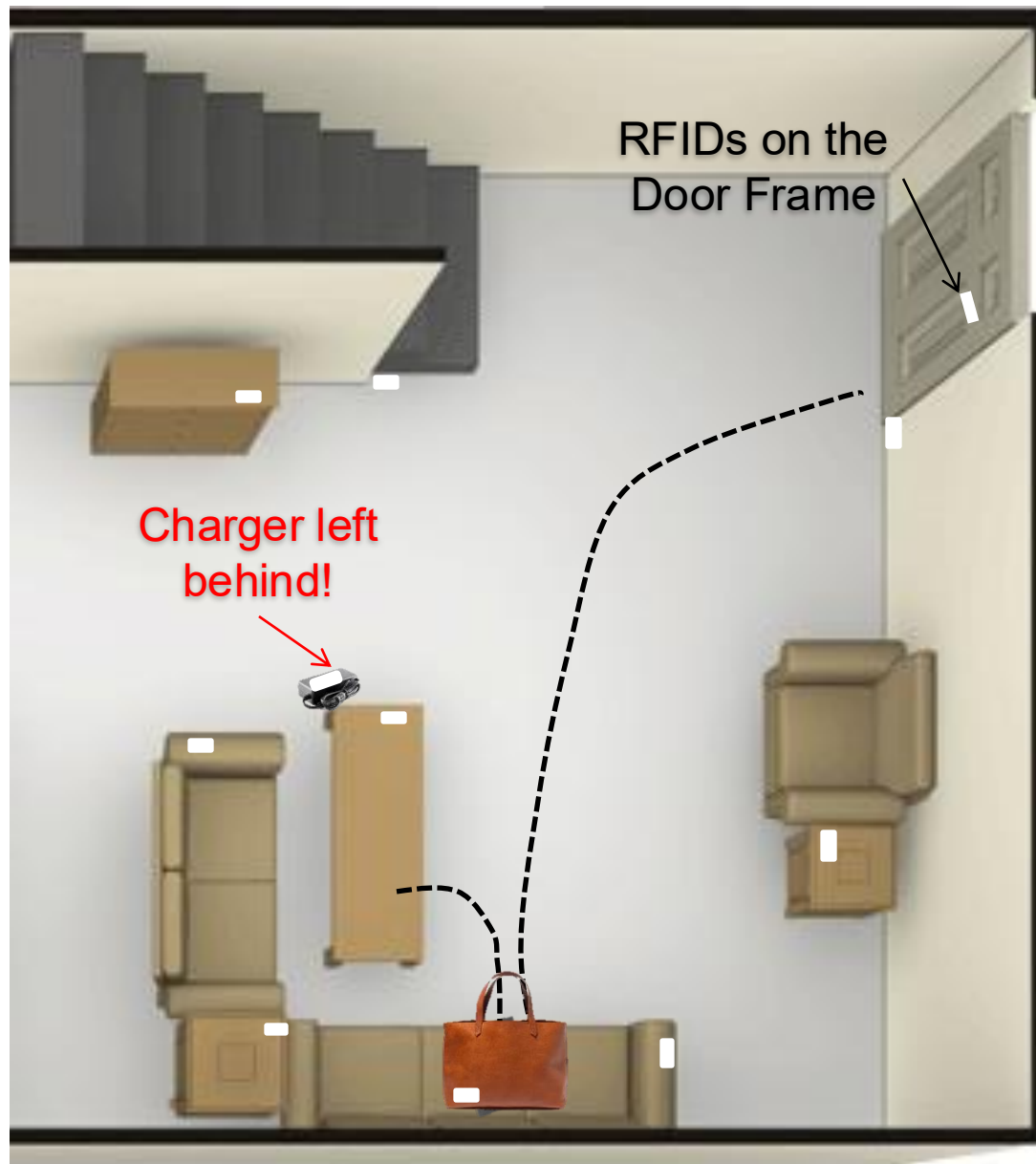
# Smart Homes



# Smart Homes



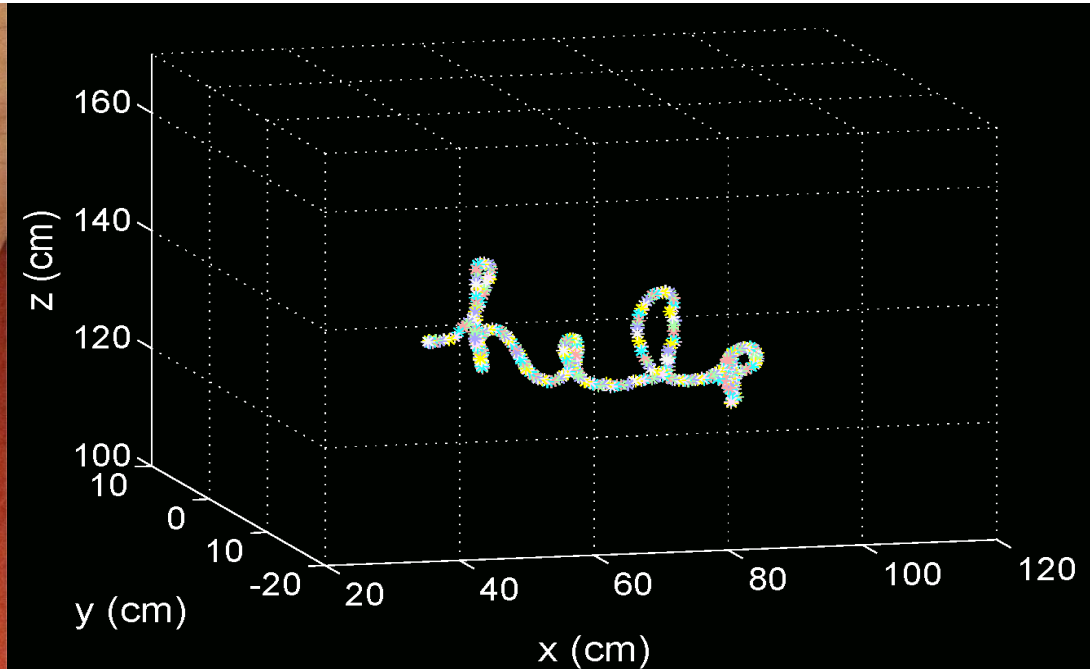
# Smart Homes



# How Do We Get Virtual Touch Screens?



# How Do We Get Virtual Touch Screens?



4 cm wide



"Clear"

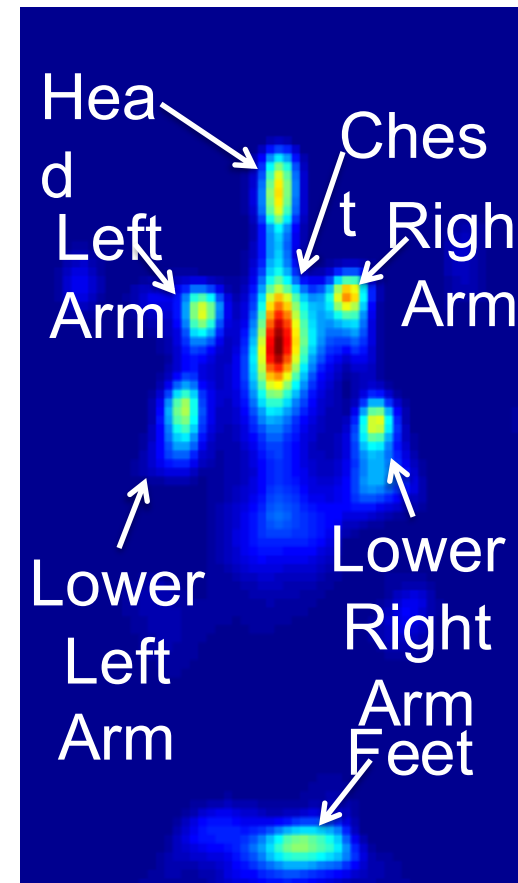
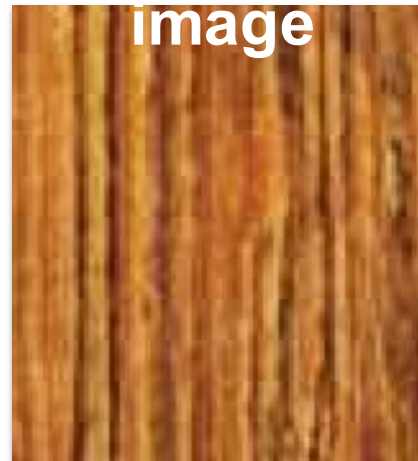
"Jue"



# Human Walks toward Sensor



# Imaging through occlusions using radio frequencies



# Baby Monitoring

2014-03-14 21:50:30



# Baby Monitoring: Breathing and Heart rate



Adib, Fadel, et al. "Smart homes that monitor breathing and heart rate." *Proceedings of the 33rd annual ACM conference on human factors in computing systems*. 2015.



# Smart homes that monitor and adapt to our breathing and heart rates?

Personal  
Health



Baby Sleep



Elderly Health



Adapt Lighting and Music to  
Mood





# Today: technologies for monitoring vital signs are cumbersome

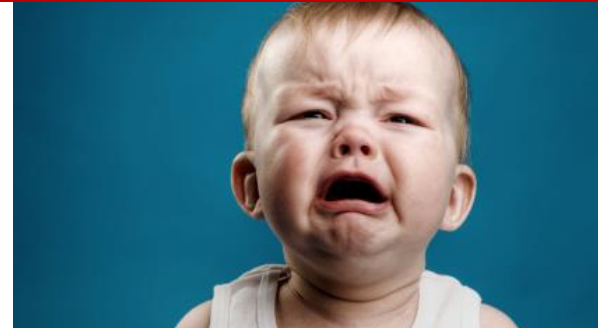
## Breath Monitoring



## Heart Rate Monitoring



**Wireless enables contactless sensing:  
sense humans without any sensors on**



# Physical Layer Security: Encryption on the Air

Encrypt using a random signal



Implant's  
signal



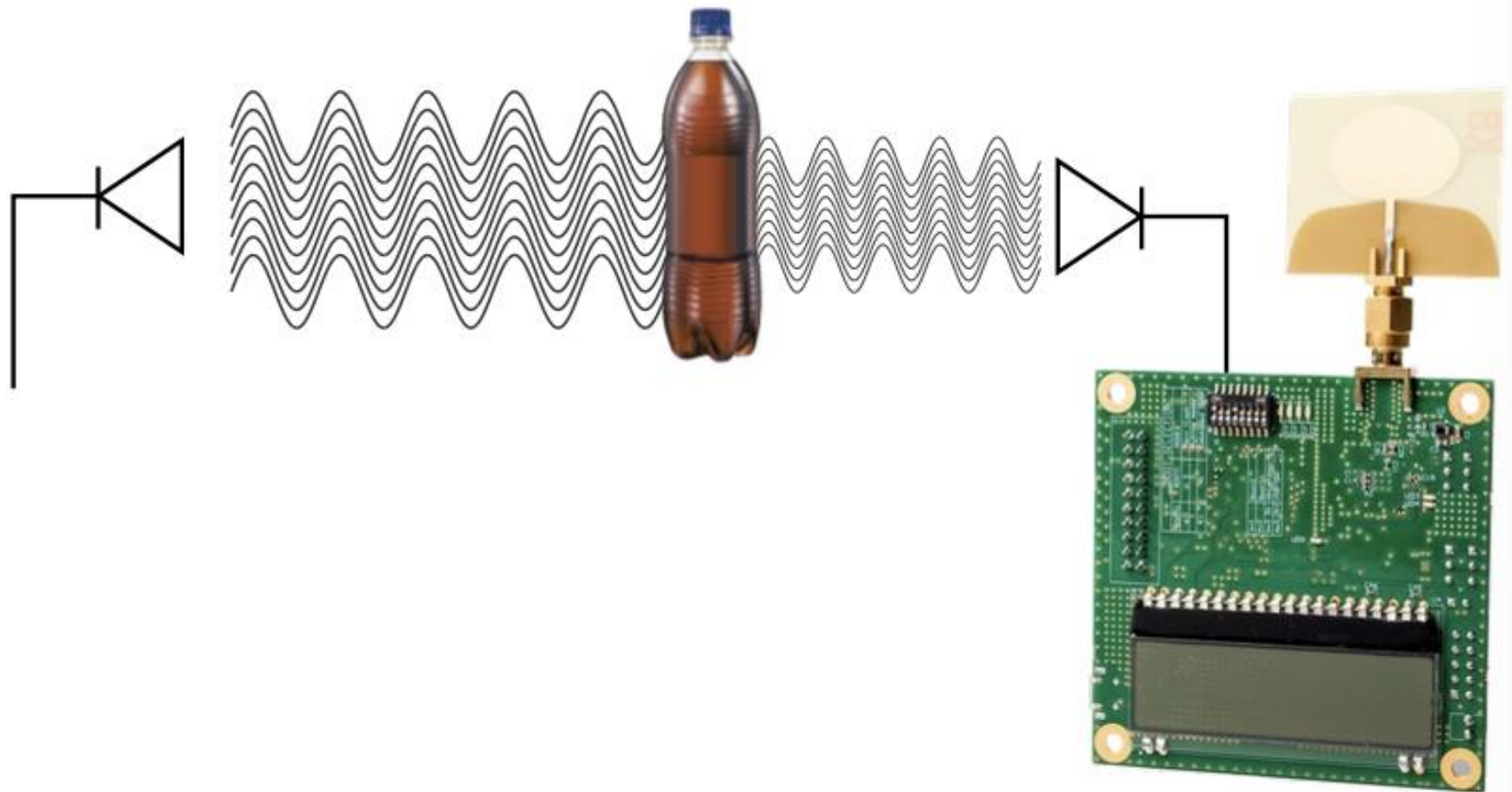
Random  
Sum



Doesn't know jamming signal

Can secure medical devices even if they have no encryption or weak encryption

# Food and Liquid Sensing



# In Body Networking & Sensing

